A Transparent Financial Risk Factor Library

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We introduce `tidyfinance`, an R package on CRAN for the core data workflow in empirical asset pricing: downloading stock returns and financial statements, constructing sorting variables, forming quantile portfolios, and computing long-short factor returns. A unified `download_data()` interface covers WRDS (CRSP, Compustat), Ken French's Data Library, FRED, and further sources. Modular functions handle fiscal-year lagging, breakpoint computation, and portfolio aggregation; every construction choice has a documented default and can be varied independently, making pipelines transparent and reproducible. We illustrate with a risk factor library covering 50 sorting variables across 187008 precomputed specifications; construction choices shift Sharpe ratios enough to reverse the sign of the premium for several factors. A programmatic API, an interactive dashboard at factors.tidy-finance.org, and `py-tidyfinance` extend access to Python researchers.

Introduction: Constructing financial risk factors

Factor returns drive empirical asset pricing tests, portfolio allocation, and performance attribution. Every series, however, embeds dozens of methodological choices: how breakpoints are set, whether portfolios are equal- or value-weighted, how delistings are handled, and which fiscal-year lag is applied. These choices are rarely standardized and almost never published with code. Researchers who reproduce or extend a factor series must reconstruct each pipeline from scratch, with no guarantee their choices match those of the original authors.

No widely adopted, transparent source standardizes factor construction. Current practice relies on informal publication of factor series without code or documented workflows, limiting reproducibility, comparability, and extension. Existing resources illustrate the gap: the French Data Library (Eugene F. Fama and French 2023) provides canonical factors but no code; the Open Source Asset Pricing Project (Chen and Zimmermann 2022) focuses on replication; the global factor database (Jensen, Kelly, and Pedersen 2023) targets Python only. No unified, open-source alternative exists.

We introduce `tidyfinance` (R, on CRAN): a modular framework for reproducible factor construction. The library covers 50 sorting variables and 187008 total specifications. We document and test every design choice (breakpoints, weighting, rebalancing, sample inclusion) and expose each as a configurable parameter.

The package implements an end-to-end pipeline: data acquisition from the Wharton Research Data Services (WRDS), Compustat, and other common providers; automated variable construction with standard lagging and cleaning routines; and portfolio aggregation with configurable sorting. A programmatic application programming interface (API) provides access to precomputed factor series with versioning and batch retrieval. An interactive dashboard at factors.tidy-finance.org makes the library accessible without code. Python researchers access the same data download functions and factor library via `py-tidyfinance`. All software and workflows are publicly available to enable replication, sensitivity analysis, and extension.

Hou, Xue, and Zhang (2020) replicate 452 cross-sectional anomalies and find that most fail to survive out-of-sample, attributing failures in part to undocumented construction variation across studies. Chen and Zimmermann (2022) respond by publishing code for 212 anomaly portfolios, but their library targets a fixed, documented set rather than a reconfigurable pipeline. Jensen, Kelly, and Pedersen (2023) provide a global factor database without an open construction workflow. `tidyfinance` fills the gap all three leave open: every construction decision is explicit, configurable, and version-controlled, so researchers can trace and vary any assumption systematically.

Among R packages for quantitative finance, tidyquant (Dancho and Vaughan 2026), PerformanceAnalytics (Peterson and Carl 2026), and quantmod (Ryan and Ulrich 2025) support time-series retrieval and portfolio performance analytics. None begins at raw accounting and return data, applies the cleaning and lagging conventions factor construction requires, or exposes construction choices as configurable parameters. `tidyfinance` provides this upstream pipeline in R with a consistent API; `py-tidyfinance` extends the same interface to Python. Researchers access standardized factor series without rebuilding data workflows in each environment.

Package overview

Design and workflow

`tidyfinance` organises the empirical asset pricing workflow into three stages: data acquisition, variable construction, and portfolio sorting. Figure 1 shows the pipeline and the package functions at each stage. Each stage exposes modular functions with documented defaults and configurable parameters. Researchers can adopt the full pipeline or call individual functions within their own workflows.

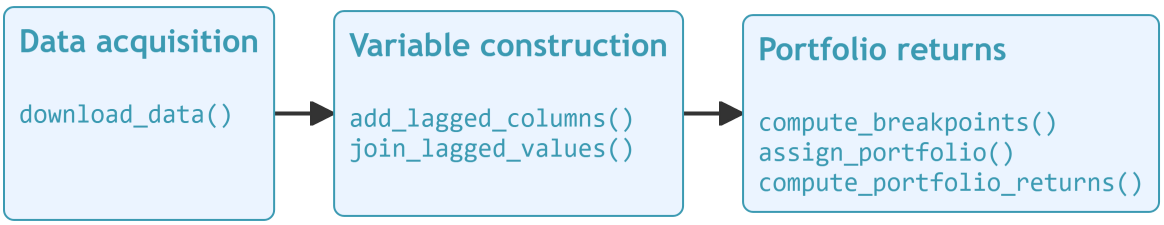


Figure 1: Research workflow for factor construction with `tidyfinance`, illustrating the three-stage pipeline and the package functions involved at each step.

Data access

`tidyfinance` provides a single interface, `download_data()`, to 319 dataset types: equity returns, fixed-income transactions, macroeconomic predictors, index constituents, and more. Table 1 lists all available sources. `list_supported_types()` and `list_supported_indexes()` return the full catalogue programmatically. The same interface is provided in Python via `tidyfinance`, so researchers can switch environments without rebuilding data workflows.

Table 1: Data sources available via `download_data()` in `tidyfinance`.

Source	Description	Example
Public data sources		
Ken French's Data Library	Fama-French 169+ factor models, portfolios, industry classifications	`download_data("famafrench", "Fama/French 5 Factors")`
Global Q	Hou-Xue-Zhang Q5 factor model (market, size, investment, ROE) at multiple frequencies	`download_data("globalq", "factors_q5_monthly")`
Goyal-Welch Predictors	Macroeconomic return predictors (DP, EP, BM, spreads, etc.)	`download_data("macro_predictors", "monthly")`
FRED	St. Louis Fed economic time series (GDP, CPI, etc.)	`download_data("fred", series = "GDP")`
Yahoo Finance	Daily OHLCV with adjusted close for any ticker	`download_data("stock_prices", symbols = "AAPL")`
Index Constituents	Current members of 21 major indexes (S&P 500, Russell, DAX, MSCI World, etc.)	`download_data("constituents", index = "DAX")`

Source	Description	Example
Open Source Asset Pricing	213+ monthly asset pricing signals and anomalies	`download_data("osap")`
WRDS (subscription required)		
CRSP	Stock returns, market cap, exchange, industry	`download_data("wrds", "crsp_monthly")`
Compustat	Annual and quarterly financial statements	`download_data("wrds", "compustat_annual")`
CCM Links	CRSP-Compustat linking table (permno to gvkey)	`download_data("wrds", "ccm_links")`
FISD	Mergent Fixed Income Securities Database	`download_data("wrds", "fisd")`
TRACE	Corporate bond transactions (Dick-Nielsen cleaned)	`download_data("wrds", "trace_enhanced", cusips = "655844CJ5")`

Standard data cleaning

`download_data()` applies standard cleaning conventions automatically. For CRSP monthly data (version 2), the function restricts the universe to common equity securities traded on NYSE, AMEX, or NASDAQ. It computes market capitalisation as shares outstanding times price, constructs a one-month lag for portfolio weights, and maps exchange codes and SIC codes to readable labels. Excess returns are raw returns minus the Fama-French risk-free rate; observations with missing returns or market capitalisation are dropped. CRSP version 2 incorporates delisting returns directly, eliminating the need for a separate adjustment.

For Compustat annual data, `download_data()` restricts to industrial-format consolidated statements and retains only the most recent observation per firm-year. It constructs book equity through a coalesce hierarchy: stockholders' equity, then common equity plus preferred stock, then total assets minus total liabilities. The function adjusts for deferred taxes and investment tax credits and subtracts preferred stock.

Vignette: A case study in factor construction

We walk through the construction of the asset growth factor from raw CRSP and Compustat data to long-short portfolio returns, implementing each stage of Figure 1 in turn. Each step uses a `tidyfinance` function and generalises directly to any sorting variable in the factor library.

Installation and setup

You can install the released version of `tidyfinance` from CRAN via:

```
install.packages("tidyfinance")
```

To install the development version of `tidyfinance` from GitHub, first install the pak package if needed, then install the development version:

```
pak::pak("tidy-finance/r-tidyfinance")
```

Once installed, load `tidyfinance` and a minimal set of packages for data manipulation and visualisation.

```
library("tidyfinance")
```

```
library("dplyr")
library("tidyr")
```

```
Warning: package 'tidyr' was built under R version 4.4.3
```

```
library("lubridate")
```

```
Warning: package 'lubridate' was built under R version 4.4.3
```

Data download

Before the first WRDS download, call `set_wrds_credentials()` to store WRDS credentials securely in `.Renviron`; subsequent `download_data()` calls establish the database connection automatically.

```
date_start <- as.Date("1960-01-01")
date_end <- as.Date("2024-12-31")
```

First, we download monthly stock returns from CRSP. Setting `add_ccm_links = TRUE` appends CRSP-Compustat linking identifiers needed to merge with financial statement data in the next step.

```
crsp_monthly <- download_data(
  domain = "wrds",
  dataset = "crsp_monthly",
  version = "v2",
  start_date = date_start,
  end_date = date_end,
  add_ccm_links = TRUE
)
```

The dataset contains 3451594 stock-month observations for 26263 unique stocks, covering the period from 1960-01-01 to 2024-12-01.

Next, we download annual financial statements from Compustat.

```
compustat_annual <- download_data(
  domain = "wrds",
  dataset = "compustat_annual",
  start_date = date_start,
  end_date = date_end,
  additional_columns = c("at", "ib"),
  only_us = TRUE
)
```

Annual Compustat data contains 506778 firm-year observations for 39587 unique firms, covering the period from 1960-01-01 to 2024-12-01. The `additional_columns` argument retrieves total assets (`at`) and income before extraordinary items (`ib`), needed for computing asset growth and applying an earnings filter in later steps. Setting `only_us = TRUE` restricts the dataset to USD-denominated observations, ensuring consistency with CRSP.

Data cleaning and variable construction

We illustrate data cleaning and variable construction using asset growth (AG) as our running example. Asset growth is the year-over-year percentage change in total assets (Cooper, Gulen, and Schill 2008),

$$AG_t = (AT_t - AT_{t-1})/AT_{t-1}.$$

The `add_lagged_columns()` function from `tidyfinance` handles temporal alignment by creating a lagged version of total assets grouped by firm identifier (`gvkey`) with a one-year lag. Non-finite values (arising when lagged total assets are zero or missing) become `NA`.[1]

[1]For the sake of simplicity we do not show every step of the data cleaning process in the paper. For instance, we apply an earnings filter to exclude firms with negative or zero income before extraordinary items (`ib`),

```r
sorting_variable_data <- compustat_annual |>
  add_lagged_columns(
    "at",
    lag = years(1),
    by = "gvkey",
  ) |>
  mutate(
    asset_growth = (at - at_lag) / at_lag,
    asset_growth = if_else(is.finite(asset_growth), asset_growth, NA_real_)
  ) |>
  select(gvkey, date, asset_growth)
```

`sorting_variable_data` contains one observation per firm-year, the natural unit for cross-sectional analysis of annual accounting variables. The package provides `filter_out()` as a convenience wrapper that removes rows matching a logical predicate, and `winsorize()` to clip numeric vectors at specified quantile bounds. Asset growth is heavily right-skewed: a small number of firms with dramatic balance-sheet expansions generate extreme positive values. We winsorize at the 1st and 99th percentiles before computing summary statistics. Table 2 reports the pooled cross-sectional summary; Table 3 shows how those moments vary across calendar years.

Table 2: Cross-sectional summary statistics for asset growth (winsorized at the 1st and 99th percentiles). The table reports the number of firm-year observations (N), the mean, standard deviation (SD), and selected percentiles of the annual asset growth distribution computed from `sorting_variable_data`.

N	Mean	SD	P10	P25	Median	P75	P90
389123	0.238	0.858	-0.191	-0.034	0.064	0.207	0.588

which is a common practice in the literature to mitigate the influence of financial distress on asset growth portfolios. This filter is implemented in the code but not shown in the paper for brevity. We also apply an age filter to exclude firms younger than two years, which is common in the literature to ensure that the sorting variable is based on stable financial information. This filter is also implemented in the code but not shown in the paper for brevity. The factor library API provides preprocessed panels with all these filters applied, so users can access the final sorting variables without having to implement these steps themselves.

Table 3: Time-series averages of annual cross-sectional summary statistics for asset growth (winsorized at the 1st and 99th percentiles). For each calendar year, cross-sectional mean, standard deviation, and median are computed from `sorting_variable_data`; the table reports the time-series mean, minimum, and maximum of those annual statistics.

Years	Avg N	Mean (avg)	Mean (min)	Mean (max)	SD (avg)	Median (avg)	Median (min)	Median (max)
64	6080	0.242	0.048	0.889	0.847	0.067	0.001	0.132

The next step merges annual sorting variables onto the monthly CRSP stock panel using `join_lagged_values()`. The function aligns fiscal-year-end data with monthly observations through a configurable lag window. Setting `ff_adjustment = TRUE` implements the Fama-French convention (Eugene F. Fama and French 1993), aligning fiscal-year data from year t with portfolio formation in July of year $t+1$ through a 12-to-18-month look-back window. The same lagging process generalizes to all sorting variables in the full factor library. The factor library API provides precomputed lagged panels for all variables.

Figure 2 contrasts the two lagging mechanisms. The upper panel shows a standard look-back window controlled by `lag` and `max_lag`. The lower panel implements the Fama-French convention (Eugene F. Fama and French 1993): fiscal-year data from year t becomes available only from July of year $t+1$, so all months from July through June share the same lagged value locked in at the previous rebalancing. In both cases, `add_lagged_columns()` selects the most recent observed value within the window.

With both lagging mechanisms documented, we apply `join_lagged_values()` to merge the sorting variable onto the monthly return panel. Figure 3 shows the 33rd, 50th, and 66th percentiles of asset growth over time in the resulting panel; pronounced spikes during economic expansions and contractions highlight the cyclical nature of corporate investment that portfolio sorts aim to exploit.

```
sorting_data <- crsp_monthly |>
  join_lagged_values(
    sorting_variable_data,
    "gvkey",
    min_lag = months(7),
    max_lag = months(18),
    ff_adjustment = TRUE
  )

sorting_data <- sorting_data |>
  select(date, permno, exchange, ret_excess, mktcap_lag, asset_growth) |>
  filter(date > date_start + months(12) + months(18))
```

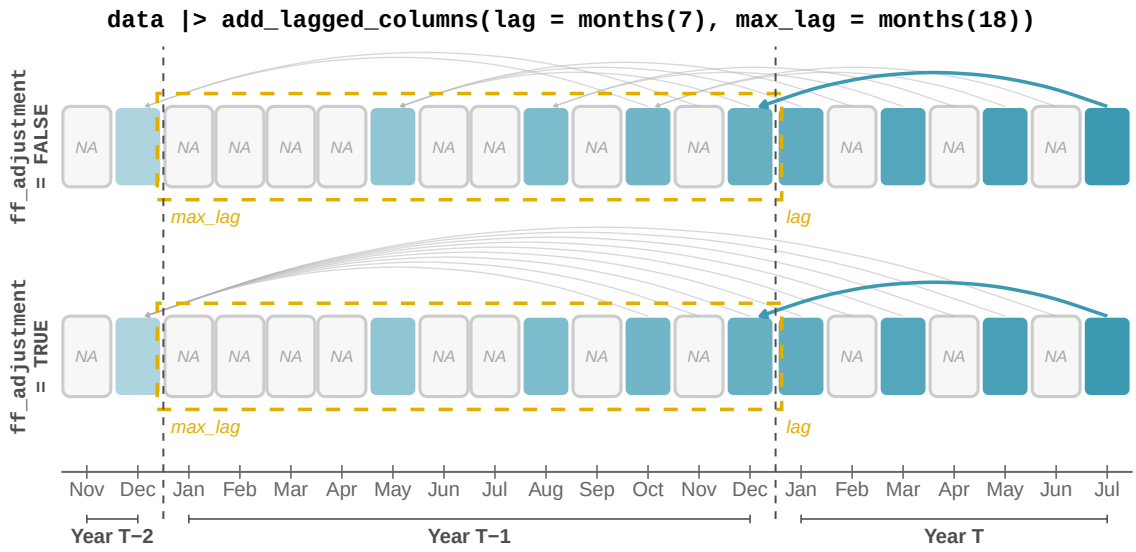


Figure 2: Two lagging mechanisms in `add_lagged_columns()`. Each row represents a data series with observed (filled) and missing (hollow) cells. Dashed rectangles mark the search window; arrows show which value gets matched to each target month.

The filter removes the initial year used to compute the growth rate and an additional 18 months to avoid mechanically introduced gaps from the lagging step, leaving a panel of stock-month observations with excess returns, exchange membership, lagged market capitalization, and the lagged sorting variable.

With the sorting variable constructed and merged onto the monthly stock panel, we now turn to portfolio sorts, the mechanism that translates firm-level characteristics into tradable factor returns.

Portfolio sorts

In portfolio sorts, stocks are ranked each period by their sorting variable and divided into quantile portfolios. Portfolio returns are equal-weighted or value-weighted averages of constituent stock returns. The long-short factor return is the difference between the extreme portfolios, for instance bottom minus top for asset growth, reflecting the negative premium documented by Cooper, Gulen, and Schill (2008). Key methodological decisions include the number of portfolios, the exchange universe for computing breakpoints, the weighting scheme, and the rebalancing frequency.

The `tidyfinance` package implements this workflow through three functions. First, `breakpoint_options()` configures the sorting parameters: the number of quantile portfolios

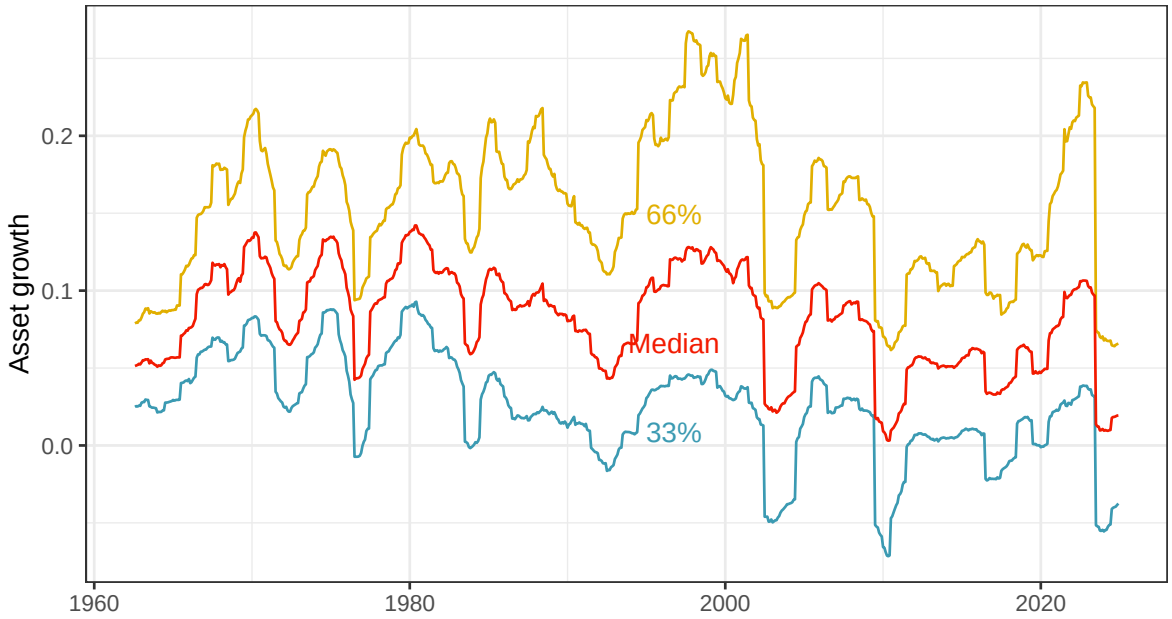


Figure 3: Distribution of asset growth over time. Lines show the 33rd, 50th, and 66th percentiles of the cross-sectional distribution of asset growth in `sorting_data`, the firm-month panel used as input for portfolio construction. Asset growth is expressed as a proportion (e.g., 0.2 = 20% annual growth in total assets) and covers the full sample period of the merged CRSP-Compustat dataset.

and which exchanges determine the breakpoints. Second, `compute_portfolio_returns()` takes the sorting data, applies the specified configuration, and returns portfolio-level returns. Third, `compute_long_short_returns()` constructs the hedge portfolio return from the extreme quantiles.

Univariate sorts

`compute_portfolio_returns()` is built on two lower-level functions that are also available for custom workflows. `compute_breakpoints()` computes quantile cutpoints for the sorting variable from a specified exchange universe. `assign_portfolio()` maps each stock-month to a portfolio number based on those cutpoints. Researchers who need finer control, for instance to build overlapping portfolios, apply non-standard cutoffs, or weight within cells differently, can call these functions directly rather than through the high-level wrapper.

In a univariate sort, stocks are ranked on a single characteristic. We illustrate this with asset growth, forming decile portfolios using NYSE breakpoints and value-weighted returns, rebalanced annually in July.

```
options_main <- breakpoint_options(
  n_portfolios = 5,
  breakpoint_exchanges = "NYSE"
)
```

The `breakpoint_options()` call specifies that breakpoints come from NYSE-listed stocks only, a common convention that avoids the influence of small AMEX and NASDAQ firms on portfolio cutoffs. We then pass these options to `compute_portfolio_returns()`.

```
portfolio_returns <- compute_portfolio_returns(
  sorting_data,
  sorting_variables = "asset_growth",
  sorting_method = "univariate",
  rebalancing_month = 7,
  options_main
)
```

```
Returning a complete panel with 55 missing values in factor returns due to
insufficient observations (n() < 0).
```

`compute_portfolio_returns()` returns a long tibble with one row per portfolio-month, containing value-weighted (`ret_excess_vw`), equal-weighted (`ret_excess_ew`), and capped value-weighted (`ret_excess_vw_capped`) excess returns. The long-short factor return is obtained via `compute_long_short_returns()`. The `direction = "bottom_minus_top"` argument means

the factor is long low-AG stocks and short high-AG stocks, consistent with the negative premium documented by Cooper, Gulen, and Schill (2008).

```
factor_returns_ag <- portfolio_returns |>
  compute_long_short_returns(direction = "bottom_minus_top")
```

Table 4 summarises the time series of each quintile across all three weighting schemes, with the long-short portfolio appended as the final row.

Table 4: Time-series averages of annualised monthly excess returns for asset growth quintile portfolios. Portfolios are formed annually in July using NYSE breakpoints, rebalanced annually. Means are annualised (multiplied by 12). The t~statistic tests whether the mean monthly excess return differs from zero. The L/S row is the long-short portfolio (bottom minus top quintile).

Portfolio	VW Mean	VW t-stat	EW Mean	EW t-stat	VWC Mean	VWC t-stat
1	0.093	4.349	0.149	4.816	0.103	4.085
2	0.079	4.343	0.117	5.139	0.095	4.530
3	0.071	3.796	0.108	4.845	0.092	4.425
4	0.073	3.677	0.103	4.297	0.088	3.935
5	0.067	2.667	0.050	1.630	0.056	1.981
L/S	0.026	1.924	0.099	9.116	0.047	4.468

The value-weighted long-short portfolio earns 2.6% per year (t~statistic 1.92), consistent with the negative premium documented by Cooper, Gulen, and Schill (2008). The equal-weighted spread is larger, as expected for characteristics that correlate with size. The monotone decline across quintiles confirms that `tidyfinance` reproduces the literature result from raw WRDS data.

Bivariate sorts

Bivariate (double) sorts control for a second variable, typically firm size (`mktcap_lag`), when forming portfolios. In dependent sorts, stocks are first sorted into groups by the secondary variable; within each group, they are then sorted on the main characteristic. In independent sorts, both sorts apply unconditionally, and portfolios form at the intersections. The factor return averages across the secondary dimension, isolating the effect of the primary variable after controlling for size. Figure 4 illustrates the three approaches for a single cross-section.

```r
cross_section <- sorting_data |>
  filter(date == "2024-12-01") |>
  mutate(
    asset_growth = rank(asset_growth) / n(),
    mktcap_lag = rank(mktcap_lag) / n()
  )

n_portfolio <- 2

breakpoints_ag <- cross_section |>
  compute_breakpoints(
    "asset_growth",
    breakpoint_options = breakpoint_options(n_portfolios = n_portfolio)
  )

portfolios_univariate <- cross_section |>
  mutate(
    ag_portfolio = assign_portfolio(
      pick(everything()),
      "asset_growth",
      breakpoint_options = breakpoint_options(n_portfolios = n_portfolio)
    )
  )

portfolios_independent <- cross_section |>
  mutate(
    ag_portfolio = assign_portfolio(
      pick(everything()),
      "asset_growth",
      breakpoint_options = breakpoint_options(n_portfolios = n_portfolio)
    ),
    size_portfolio = assign_portfolio(
      pick(everything()),
      "mktcap_lag",
      breakpoint_options = breakpoint_options(n_portfolios = n_portfolio)
    ),
    portfolio_cell = paste0(ag_portfolio, "-", size_portfolio)
  )

portfolios_dependent <- cross_section |>
  mutate(
    ag_portfolio = assign_portfolio(
```

```
      pick(everything()),
      "asset_growth",
      breakpoint_options = breakpoint_options(n_portfolios = n_portfolio)
    )
  ) |>
  group_by(ag_portfolio) |>
  mutate(
    size_portfolio = assign_portfolio(
      pick(everything()),
      "mktcap_lag",
      breakpoint_options = breakpoint_options(n_portfolios = n_portfolio)
    ),
    portfolio_cell = paste0(ag_portfolio, "-", size_portfolio)
  ) |>
  ungroup()
```

We reuse `options_main` from the univariate sort (five portfolios, NYSE breakpoints) and add a separate options object for the secondary size dimension:

```
options_secondary <- breakpoint_options(
  n_portfolios = 2,
  breakpoint_exchanges = "NYSE"
)

portfolio_returns_biv <- compute_portfolio_returns(
  sorting_data,
  c("asset_growth", "mktcap_lag"),
  sorting_method = "bivariate-dependent",
  rebalancing_month = 7,
  options_main,
  options_secondary
)
```

```
Returning a complete panel with 55 missing values in factor returns due to
insufficient observations (n() < 0).
```

```
factor_returns_biv <- portfolio_returns_biv |>
  compute_long_short_returns(direction = "bottom_minus_top")
```

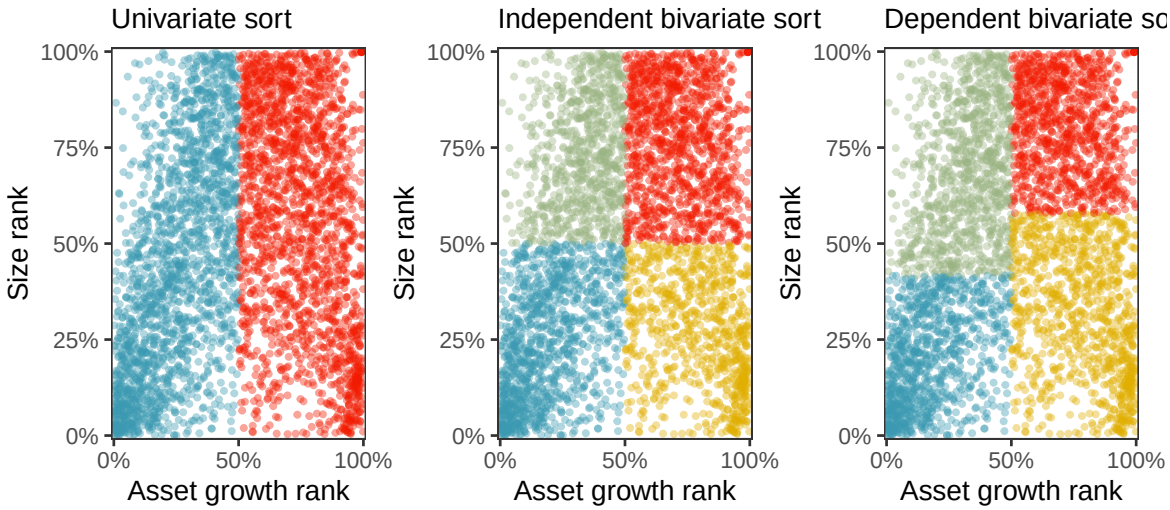


Figure 4: Portfolio sort approaches illustrated for a single month (December 2024). Each panel plots stock-month observations assigned to `r .QuartoInlineRender(n_portfolio)`×`r .QuartoInlineRender(n_portfolio)` portfolios based on asset growth rank (x-axis) and size rank (y-axis), both expressed as percentiles. Colors indicate portfolio cell assignment. The univariate sort (left) assigns portfolios on asset growth alone; the independent bivariate sort (center) assigns each dimension independently; the dependent bivariate sort (right) sorts on size first, then on asset growth within size groups.

Table 5: Univariate vs size-controlled long-short returns for asset growth. Annualized value-weighted mean excess return and t~statistic for the univariate sort and the dependent bivariate sort averaging across two size groups. Portfolios use NYSE breakpoints and annual July rebalancing.

Sort	VW Mean	VW t~stat
Univariate	0.026	1.924
Bivariate (size-controlled)	0.036	3.376

The AG premium survives size control, suggesting it captures an investment effect distinct from the size factor. The `sorting_method` argument switches between dependent and independent double sorts, so researchers can inspect both configurations without modifying the pipeline.

Factor library and API

Specification grid

The examples above represent single points in a high-dimensional space of methodological choices. Table 6 summarises the preprocessing decisions that can be varied when constructing factor returns. The full factor library exhaustively evaluates all valid combinations of these choices for each of the sorting variables listed in Table 7.

Table 6: Preprocessing choices in the portfolio sort grid and corresponding `download_data()` arguments. Italic values indicate the default. For double sorts, the secondary quantiles and exchange choices apply to the size dimension only; for univariate sorts, those rows are not applicable. FF refers to the lag procedure of Eugene F. Fama and French (1993).

Preprocessing choice	Argument	Options	Description
Size exclusion	`exclude_size`	None; *bottom 20%*	Fraction of the smallest stocks (by NYSE market cap) excluded from the universe.
Exclusion of financials	`exclude_financials`	*No*; Yes	Drops financial-sector stocks (SIC 6000–6999) from the universe.
Exclusion of utilities	`exclude_utilities`	*No*; Yes	Drops utility-sector stocks (SIC 4900–4999) from the universe.
Exclusion of neg. earnings	`exclude_negative_earnings`	*No*; Yes	Drops firms with negative earnings before sorting.

Table 6: Preprocessing choices in the portfolio sort grid and corresponding `download_data()` arguments. Italic values indicate the default. For double sorts, the secondary quantiles and exchange choices apply to the size dimension only; for univariate sorts, those rows are not applicable. FF refers to the lag procedure of Eugene F. Fama and French (1993).

Preprocessing choice	Argument	Options	Description
Sorting variable lag	`sorting_variable_lag`	3; 1; *6m*; FF	Lag applied to the sorting variable before portfolio assignment.
Rebalancing	`rebalancing`	*Monthly*; Annual	How frequently portfolios are reformed.
BP: Quantiles (main)	`breakpoints_main`	3; 5; *10*	Number of quantile groups for the portfolio sort.
Sort method	`sorting_method`	*Single*; Double dependent; Double independent	Single sort or sequential double sort (by size).
BP: Quantiles (secondary)	`breakpoints_secondary`	*NA*; 2; 5	Groups for the secondary sort variable; only relevant for double sorts.
BP: Exchanges	`breakpoints_exchanges`	*NYSE*; All	Exchange(s) used to compute quantile breakpoints.
Weighting scheme	`weighting_scheme`	EW; *VW*; truncated VW	Return weighting within portfolios.

We host precomputed factor returns for all specifications on Hugging Face and make them accessible via the `tidyfinance` API. Researchers can retrieve results without re-running the computationally intensive portfolio sort pipeline. An interactive download interface is also available at factors.tidy-finance.org.

The factor library covers 50 sorting variables across seven categories: investment, profitability, valuation, momentum, intangibles, trading frictions, and financing. The selection follows Jensen, Kelly, and Pedersen (2023). Table 7 lists all variables with their long-short direction and primary reference.

Construction-choice sensitivity

The vignette above built one specification of the AG factor. The library evaluates every valid combination of the choices in Table 6 for each sorting variable. Figure 5 shows how the AG Sharpe ratio varies by lag convention and weighting scheme across those specifications.

```
Warning: Removed 189 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

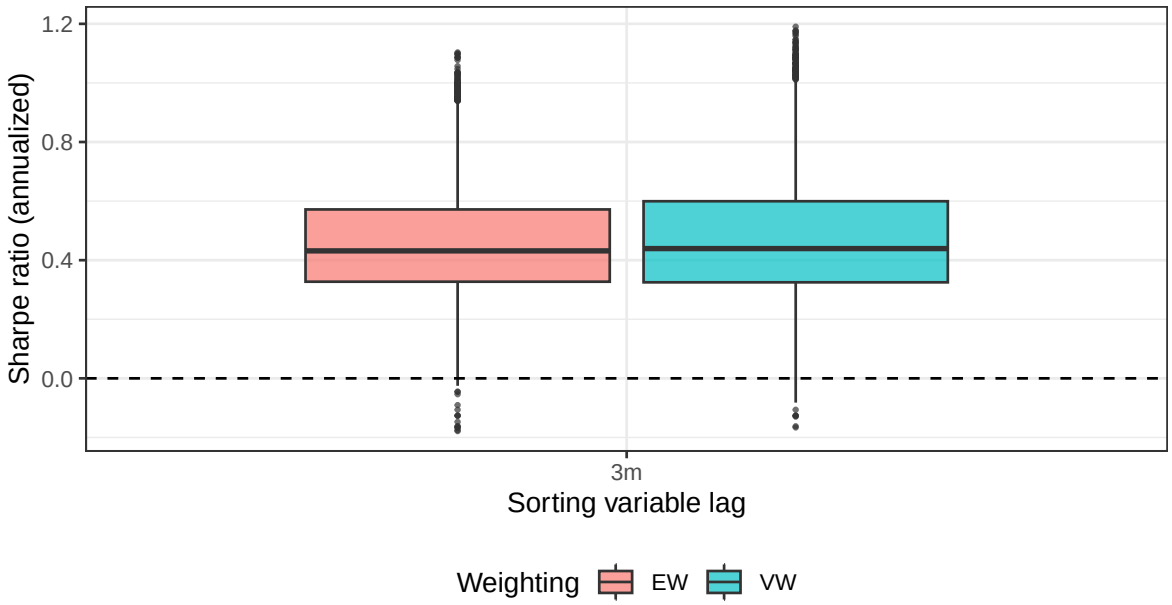


Figure 5: AG Sharpe ratio by lag convention and weighting scheme. Each observation is one specification from the full grid (Table 6); remaining choices vary across all valid combinations. The dashed line marks zero.

The factor library

This sensitivity is not specific to AG. For all 50 sorting variables in the library, construction choices shift Sharpe ratios by enough to reverse the sign of the premium for some factors. Figure 6 shows the distribution of annualized Sharpe ratios for each variable across all 187008 specifications.

Figure 6 conveys two findings. First, most distributions are wide: construction choices have material effects on factor performance across the library. Second, for several factors the distribution spans zero, meaning the premium's sign depends on the specification. A researcher reporting a single undocumented series risks mistaking a construction artifact for an economic signal. Factors with narrow distributions centered well above zero are construction-robust; those with wide or zero-straddling distributions require sensitivity analysis before drawing conclusions.

Table 7 lists all 50 sorting variables with their long-short direction and primary reference.

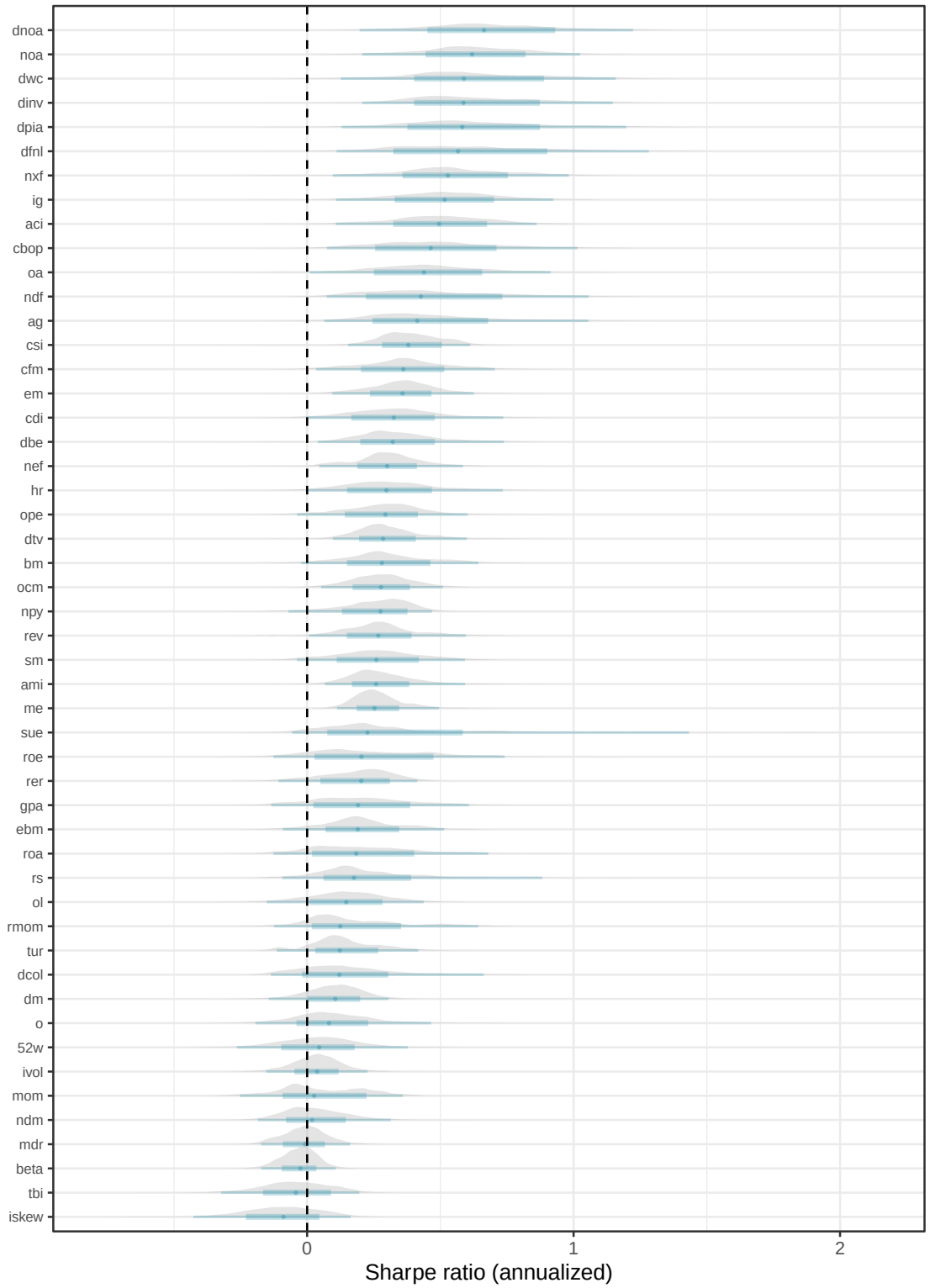


Figure 6: Sharpe ratios across construction choices. Ridge plots show the distribution of annualized Sharpe ratios of long-short return differentials for each sorting variable across all `r .QuartoInlineRender(nrow(portfolio_sort_grid))` specifications illustrated in Table 6, based on the available sample period per factor. Most variables exhibit substantial variation across specifications, underscoring the sensitivity of factor returns to construction choices.

Table 7: Sorting variables in the factor library. The direction column indicates whether the long-short factor return goes long top-minus-bottom (↑) or bottom-minus-top (↓) of the characteristic distribution.

Variable	Name		Main reference
sv_52w	52-Week High	$\uparrow$	\cite{george.2004}
sv_aci	Abnormal Corporate Investment	$\downarrow$	\cite{titman.2004}
sv_ami	Amihud Illiquidity	$\uparrow$	\cite{amihud.2002}
sv_ag	Asset Growth	$\downarrow$	\cite{cooper.2008}
sv_beta	Beta	$\uparrow$	\cite{fama.1973}
sv_bm	Book-to-Market	$\uparrow$	\cite{daniel.2006}
sv_cfm	Cash Flow to Market	$\uparrow$	\cite{lakonishok.1994}
sv_cbop	Cash-Based Operating Profitability	$\uparrow$	\cite{ball.2016}
sv_dbe	Change in Common Equity	$\downarrow$	\cite{richardson.2005}
sv_dcol	Change in Current Operating Liabilities	$\downarrow$	\cite{richardson.2005}
sv_dfnl	Change in Financial Liabilities	$\downarrow$	\cite{richardson.2005}
sv_dnoa	Change in Net Operating Assets	$\downarrow$	\cite{hirshleifer.2004}
sv_dpia	Change in PPE and Intangibles	$\downarrow$	\cite{lyandres.2008}
sv_dwc	Change in Working Capital	$\downarrow$	\cite{richardson.2005}
sv_cdi	Composite Debt Issuance	$\downarrow$	\cite{lyandres.2008}
sv_csi	Composite Share Issuance	$\downarrow$	\cite{daniel.2006}
sv_dm	Debt to Market	$\uparrow$	\cite{bhandari.1988}
sv_dtv	Dollar Trading Volume	$\downarrow$	\cite{brennan.1998}
sv_em	Earnings to Market	$\uparrow$	\cite{basu.1983}
sv_ebm	Enterprise Book to Market	$\uparrow$	\cite{penman.2007}
sv_gpa	Gross Profit Asset	$\uparrow$	\cite{novy-marx.2013}
sv_hr	Hiring Rate	$\downarrow$	\cite{belo.2012}
sv_iskew	Idiosyncratic Skewness	$\downarrow$	\cite{bali.2016}
sv_ivol	Idiosyncratic Volatility	$\downarrow$	\cite{ang.2006}
sv_dinv	Inventory Changes	$\downarrow$	\cite{thomas.2002}
sv_ig	Investment Growth	$\downarrow$	\cite{xing.2008}
sv_rev	Long-Term Reversal	$\downarrow$	\cite{debondt.1985}
sv_me	Market Equity	$\downarrow$	\cite{banz.1981}
sv_mdr	Maximum Daily Return	$\downarrow$	\cite{bali.2011}

sv_mom	Momentum	$\uparrow$	\cite{fama.1996}
sv_ndf	Net Debt Financing	$\downarrow$	\cite{baker.2006}
sv_ndm	Net Debt to Market	$\downarrow$	\cite{penman.2007}
sv_nef	Net Equity Financing	$\downarrow$	\cite{baker.2006}
sv_nxf	Net External Financing	$\downarrow$	\cite{baker.2006}
sv_noa	Net Operating Assets	$\downarrow$	\cite{hirshleifer.2004}
sv_npy	Net Payout Yield	$\uparrow$	\cite{boudoukh.2007}
sv_o	O-Score	$\downarrow$	\cite{ohlson.1980,dichev.1998}
sv_oa	Operating Accruals	$\downarrow$	\cite{sloan.1996}
sv_ocm	Operating Cash Flow to Market	$\uparrow$	\cite{desai.2004}
sv_ope	Operating Efficiency	$\uparrow$	\cite{fama.2015}
sv_ol	Operating Leverage	$\uparrow$	\cite{novy-marx.2011}
sv_rer	Real-Estate Ratio	$\uparrow$	\cite{titman.2004}
sv_rs	Relative Strength	$\uparrow$	\cite{jegadeesh.2006}
sv_rmom	Residual Momentum	$\uparrow$	\cite{blitz.2011}
sv_roa	Return on Assets	$\uparrow$	\cite{balakrishnan.2010}
sv_roe	Return on Equity	$\uparrow$	\cite{hou.2014}
sv_sm	Sales to Market	$\uparrow$	\cite{barbee.1996}
sv_tur	Share Turnover	$\downarrow$	\cite{datar.1998}
sv_sue	Standardized Unexpected Earnings	$\uparrow$	\cite{foster.1984}
sv_tbi	Taxable Income to Book Income	$\downarrow$	\cite{lev.2004}

To download factor returns from the factor library, call

```
asset_growth_returns <- download_data(
  "tidyfinance",
  "factor_library",
  sorting_variable = "ag",
  sorting_variable_lag = "6m"
)
```

The defaults passed to `dataset = "factor_library"` reflect one common construction choice but may not suit every research question. Always verify that the selected combination matches the intended design before drawing conclusions.

```
download_data(
  "tidyfinance",
```

```
  "factor_library",
  sorting_variable = "52w",
  sorting_variable_lag = "6m",
  fill_all = TRUE
)
```

For finer control, `tidyfinance` exposes lower-level helpers that sit beneath the `download_data()` interface. `get_available_huggingface_files()` returns all available files for a given Hugging Face repository as a tibble; `download_factor_library_returns_ids()` retrieves returns for a specific vector of precomputed specification IDs. Researchers who want to iterate over a custom subset of specifications rather than filter by parameter value will prefer this route.

```
full_grid <- get_available_huggingface_files(
  "tidy-finance",
  "factor-library-grid"
) |>
  dplyr::pull(.data$url) |>
  arrow::read_parquet()

download_factor_library_returns_ids(c(123, 456))
```

Summary and discussion

Factor returns encode their construction history. Two researchers following the same paper can produce different factor series, with no way of knowing why, because breakpoints, weighting, and lagging conventions are rarely published alongside the data. `tidyfinance` makes these decisions visible: the package covers 50 sorting variables across 187008 documented specifications, each reproducible from raw WRDS and Compustat data in a single, version-controlled workflow.

The library currently covers US equities. Construction-choice sensitivity in international markets is beyond its current scope.

The broader implication is methodological. Researchers who disagree about a construction choice can resolve the disagreement empirically: run both variants through the same pipeline, compare factor returns, and report the difference. The API and interactive dashboard extend this accessibility to practitioners who need standardized series without running the full pipeline.

With 187008 specifications across 50 sorting variables, the library makes systematic sensitivity analysis newly tractable. This opens a path toward meta-analytic evidence on which factor properties are robust and which are artifacts of undocumented defaults.

Computational details

We obtained the results in this paper using R~4.5.2 with the following packages: tidyfinance~0.4.5.9022, dplyr~1.2.0, tidyr~1.3.1, ggplot2~4.0.0, lubridate~1.9.4, arrow~21.0.0.1, stringr~1.5.2, forcats~1.0.0, ggdist~3.3.3, and knitr~1.51. CRAN hosts R and all packages at https://CRAN.R-project.org/. `renv`~1.1.5 ensures full reproducibility of the computational environment. Reproducing the factor return data requires an institutional subscription to Wharton Research Data Services (WRDS), as the underlying CRSP and Compustat data are proprietary and cannot be redistributed.

We publish all code, data workflows, and documentation at https://github.com/tidy-finance/jss-multilingual-factor-library. The code is released under the MIT License, which is compatible with the GNU General Public License (GPL).

```
sessionInfo()
```

```
R version 4.4.2 (2024-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 22631)

Matrix products: default


locale:
[1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
[3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
[5] LC_TIME=German_Germany.utf8

time zone: Europe/Berlin
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices datasets  utils     methods   base

other attached packages:
[1] lubridate_1.9.5        tidyr_1.3.2            dplyr_1.2.1
[4] tidyfinance_0.4.5.9021 patchwork_1.3.2        ggforce_0.5.0
[7] ggplot2_4.0.2

loaded via a namespace (and not attached):
 [1] ggdist_3.3.3         sandwich_3.1-1       rappdirs_0.3.4
 [4] generics_0.1.4       renv_1.1.5           stringi_1.8.7
 [7] lattice_0.22-6       hms_1.1.4            digest_0.6.39
```

```
[10] magrittr_2.0.5       evaluate_1.0.5       grid_4.4.2
[13] timechange_0.4.0     RColorBrewer_1.1-3   fastmap_1.2.0
[16] blob_1.3.0           jsonlite_2.0.0       RPostgres_1.4.10
[19] DBI_1.3.0            purrr_1.2.1          scales_1.4.0
[22] frenchdata_0.2.0     tweenr_2.0.3         httr2_1.2.2
[25] cli_3.6.5            rlang_1.1.7          dbplyr_2.5.2
[28] polyclip_1.10-7      bit64_4.6.0-1        withr_3.0.2
[31] yaml_2.3.12          tools_4.4.2          tzdb_0.5.0
[34] curl_7.0.0           assertthat_0.2.1     vctrs_0.7.2
[37] R6_2.6.1             zoo_1.8-15           lifecycle_1.0.5
[40] stringr_1.6.0        bit_4.6.0            MASS_7.3-61
[43] arrow_23.0.1.2       pkgconfig_2.0.3      pillar_1.11.1
[46] gtable_0.3.6         Rcpp_1.1.1-1         glue_1.8.0
[49] xfun_0.57            tibble_3.3.1         tidyselect_1.2.1
[52] knitr_1.51           farver_2.1.2         htmltools_0.5.9
[55] labeling_0.4.3       rmarkdown_2.31       compiler_4.4.2
[58] S7_0.2.1             distributional_0.7.0
```

Acknowledgments

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Internet Appendix

Variable construction

In this section, we show the construction details for the risk factors. We group them by their underlying mechanism as shown in Table 7 in the Appendix of the main paper.

Financing

CDI - Composite Debt Issuance

Following Lyandres, Sun, and Zhang (2008), we compute composite debt issuance as the logarithmic growth rate of book debt (i.e., current debt, DLC, plus long-term debt, DLTT) over the last five years, i.e.,

$$\text{CDI}_t = \log(\text{DLC}_t + \text{DLTT}_t) - \log(\text{DLC}_{t-60} + \text{DLTT}_{t-60})$$

We replace the missing values of DLC and DLTT with zero individually. Additionally, the sum of DLC and DLTT must be larger than zero, and is set to missing otherwise.

CSI - Composite Share Issuance

Following Daniel and Titman (2006), we measure composite share issuance from CRSP data as the difference between the change in market equity (ME) (i.e., the shares outstanding, shrout, times the absolute value of their price, altprc, in million USD) and the cumulative log return (where r is a simple return) in each month from t to $t-60$, i.e.,

$$\text{CSI}_t = \log(ME_t/ME_{t-60}) - \sum_{\tau=t-59}^{t} \log(1+r_\tau)$$

DBE - Change in Common Equity

Following Richardson et al. (2005), we calculate the change in common equity as the change in common equity (CEQ) over one year scaled by lagged assets (AT), i.e.,

$$DBE_t = \frac{CEQ_t - CEQ_{t-12}}{AT_{t-12}}$$

DCOL - Change in Current Operating Liabilities

Following Richardson et al. (2005), we measure the change in current operating liabilities as the change in current operating liabilities (i.e., current liabilities, LCT, minus current debt, DLC) over one year scaled by lagged assets (AT), i.e.,

$$DCOL_t = \frac{(LCT_t - DLC_t) - (LCT_{t-12} - DLC_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC with 0.

DFNL - Change in Financial Liabilities

Following Richardson et al. (2005), we measure the change in financial liabilities (i.e., the sum of current debt, DLC, plus long-term debt, DLTT, plus preferred stocks, PSTK) over one year scaled by lagged assets (AT), i.e.,

$$DFNL_t = \frac{(DLC_t + DLTT_t + PSTK_t) - (DLC_{t-12} + DLTT_{t-12} + PSTK_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC, DLTT and PSTK with 0, if at least one of them is given.

NDF - Net Debt Financing

Following Bradshaw, Richardson, and Sloan (2006), we measure net debt financing as cash proceeds from the issuance of long-term debt (DLTIS) minus cash payments for long-term debt reductions (DLTR) plus net changes in current debt (DLCCH) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NDF_t = \frac{DLTIS_t - DLTR_t + DLCCH_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

We replace the missing values of DLCCH with zero.

NEF - Net Equity Financing

Following Bradshaw, Richardson, and Sloan (2006), we measure net equity financing as proceeds from the sale of common and preferred stocks (SSTK) minus payments for the repurchase of common and preferred stocks (PRSTKC) minus cash payments for dividends (DV) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NEF_t = \frac{SSTK_t - PRSTKC_t - DV_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

NXF - Net External Financing

Following Bradshaw, Richardson, and Sloan (2006), we measure net external financing as the sum of net debt financing (NDF) and net equity financing (NEF) as defined above.

Intangibles

HR - Hiring Rate

Following Belo, Lin, and Bazdresch (2014), we measure the hiring rate as the change in the number of employees (EMP) over one year scaled by the average number of employees, i.e.,

$$HR_t = \frac{EMP_t - EMP_{t-12}}{\frac{1}{2}(EMP_t + EMP_{t-12})}$$

We exclude firms with a hiring rate of zero.

OL - Operating Leverage

Following Novy-Marx (2011), we measure operating leverage as cost of goods sold (COGS) plus selling, general, and administrative expenses (XSGA), both scaled by current total assets (AT), i.e.,

$$OL_t = \frac{COGS_t + XSGA_t}{AT_t}$$

RER - Real-Estate Ratio

Following Tuzel (2010), we measure the real-estate ratio in two subperiods differently.

(a) Prior to 1983, the real-estate ratio is the sum of buildings (PPENB) and capital leases (PPENLS) scaled by net property, plant, and equipment (PPENT), i.e.,

$$RER_t = \frac{PPENB_t + PPENLS_t}{PPENT_t}$$

(b) After the end of 1983, the real-estate ratio is the sum of buildings at cost (FATB) and leases at cost (FATL), both divided by gross property, plant, and equipment (PPEGT), i.e.,

$$RER_t = \frac{FATB_t + FATL_t}{PPEGT_t}$$

Subsequently, we winsorize the real-estate ratios in each fiscal year at the first and 99th percentile. The industry-adjusted real-estate ratio is obtained by subtracting the industry average real-estate ratio from each stock-specific real-estate ratio. We use 2-digit SIC codes to assign stocks to industries. We always require at least five observations to calculate the industry average each year.

Investment

ACI - Abnormal Corporate Investment

Following Titman, Wei, and Xie (2004), we measure abnormal corporate investments using the variable CE, which is capital expenditures (CAPX) scaled by sales (SALE), then ACI is given by CE scaled by CE from the previous three years minus one, i.e.,

$$ACI_t = \frac{CE_t}{\frac{1}{3}(CE_{t-12} + CE_{t-24} + CE_{t-36})} - 1$$

We follow Hou, Xue, and Zhang (2020) and exclude stocks with sales below 10 million dollars.

AG - Asset Growth

Following Cooper, Gulen, and Schill (2008), we measure asset growth as the change in total assets (AT) over one year scaled by lagged assets (AT), i.e.,

$$AG_t = \frac{AT_t - AT_{t-12}}{AT_{t-12}}$$

DINV - Inventory Changes

Following Thomas and Zhang (2002), we measure the change in inventory as the annual change in inventories (INVT) scaled by average total assets (AT) over the current and the last year, i.e.,

$$DINV_t = \frac{INVT_t - INVT_{t-12}}{\frac{1}{2}(AT_t + AT_{t-12})}$$

DNOA - Change in Net Operating Assets

Following Hirshleifer et al. (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, the change in net operating assets is the difference between net operating assets of fiscal year t and fiscal year $t-12$ scaled by lagged assets (AT), i.e.,

$$DNOA_t = \frac{NOA_t^{int} - NOA_{t-12}^{int}}{AT_{t-12}}$$

DPIA - Change in Property, Plant, Equipment, and Inventory

Following Lyandres, Sun, and Zhang (2008), we measure the change in property, plant, equipment, and inventory by combining the change in gross property, plant, and equipment (PPEGT) with the annual change in inventory (INVT) and scaling this sum by lagged total assets (AT), i.e.,

$$DPIA_t = \frac{(PPEGT_t - PPEGT_{t-12}) + (INVT_t - INVT_{t-12})}{AT_{t-12}}$$

DWC - Change in Net Non-Cash Working Capital

Following Richardson et al. (2005), we measure non-cash working capital as

$$WC_t = (ACT_t - CHE_t) - (LCT_t - DLC_t)$$

where ACT corresponds to current assets, CHE to cash, LCT to current liabilities, and DLC to short-term debt. We set missing values of DLC to zero. Then, the change in net non-cash working capital corresponds to the change in WC from fiscal year $t-12$ to fiscal year t scaled by lagged total assets (AT), i.e.,

$$DWC_t = \frac{WC_t - WC_{t-12}}{AT_{t-12}}$$

IG - Investment Growth

Following Xing (2008), we measure investment growth as the annual change in capital expenditures (CAPX) scaled by lagged capital expenditures, i.e.,

$$IG_t = \frac{CAPX_t - CAPX_{t-12}}{CAPX_{t-12}}$$

NOA - Net Operating Assets

Following Hirshleifer et al. (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, net operating assets is the net operating assets of fiscal year t scaled by lagged assets (AT), i.e.,

$$NOA_t = \frac{NOA_t^{int}}{AT_{t-12}}$$

OA - Operating Accruals

Following Sloan (1996) (before 1988) and Hribar and Collins (2002) (after 1988), we measure operating accruals until 1988 as

$$OA_t = \frac{(\Delta ACT_t - \Delta CHE_t) - (\Delta LCT_t - \Delta DLC_t - \Delta TXP_t) - DP_t}{AT_{t-12}}$$

where ACT is current assets, CHE cash, LCT current liabilities, DLC short-term debt, TXP taxes payable, and DP depreciation and amortization. Moreover, we replace the missing values of DLC and TXP with zero. Δ represents changes over one year. From 1988 onwards, we measure operating accruals as

$$OA_t = \frac{NI_t - OANCF_t}{AT_{t-12}}$$

where NI is net income and OANCF corresponds to net cash flow from operations.

Momentum

MOM - Return Momentum

Following Eugene F. Fama and French (1996), we measure return momentum from monthly CRSP returns (where r is a simple return) as the cumulative return from month $t-11$ to month $t-1$ skipping the most recent month, i.e.,

$$\text{MOM}_t = \exp\left(\sum_{\tau=t-11}^{t-1} \log(1+r_\tau)\right) - 1$$

RMOM - Residual Momentum

Following Blitz, Huij, and Martens (2011), we define the 11-month residual momentum (RMOM) in each month and for each stock as cumulative residual returns from month $t-11$ to month $t-1$, scaled by the standard deviation of residual returns over the same time horizon. Residual returns ϵ are obtained in each month from regressing monthly simple, excess stock returns from month $t-1$ to month $t-36$ on the Fama and French (1993) three-factor model. Throughout these rolling regressions, we always require 36 monthly returns.

RS - Revenue Surprise

Following Jegadeesh and Livnat (2006), we measure the change in revenues per share (DRPS) from quarterly Compustat data as quarterly sales (SALEQ) scaled by the product of quarterly shares outstanding (CSHPRQ) times their cumulative adjustment factor (AJEXQ), i.e.,

$$DRPS_q = \frac{SALEQ_q}{CSHPRQ_q \cdot AJEXQ_q} - \frac{SALEQ_{q-4}}{CSHPRQ_{q-4} \cdot AJEXQ_{q-4}}$$

Then, the revenue surprise corresponds to the change in revenues per share over the last four quarters scaled by the standard deviation of the change in revenues per share over the last eight quarters. We require at least six quarterly observations for this rolling standard deviation and remove infinite values. Finally, the earnings announcement date has to be after the fiscal quarter end date.[2]

SUE - Standardized Unexpected Earnings

Following Foster, Olsen, and Shevlin (1984), we measure unexpected earnings per share (UES) from quarterly Compustat data as the change in split-adjusted earnings (which are equal to quarterly earnings per share, EPSPXQ, scaled by the correction factor for shares outstanding, AJEXQ) per share from its value four quarters ago

$$UES_q = \frac{EPSPXQ_q}{AJEXQ_q} - \frac{EPSPXQ_{q-4}}{AJEXQ_{q-4}}$$

Then, standardized unexpected earnings are defined as unexpected earnings per share divided by the standard deviation of unexpected earnings per share over the previous eight quarters. We require at least six quarterly observations for this rolling standard deviation. Finally, the earnings announcement date has to be after the fiscal quarter end date.[3]

52W - 52-Week High

Following George and Hwang (2004), we measure the 52-week high as month t's closing split-adjusted stock price scaled by the highest daily split-adjusted stock price over the previous twelve months. We only consider stock-month pairs with more than three valid price observations.

REV - Long-Term Reversal

Following De Bondt and Thaler (1985), we measure the long-term reversal effect by the cumulative returns from month $t-60$ to month $t-13$, i.e.,

[2]We follow the convention of Hou, Xue, and Zhang (2020) and do not estimate a random walk model with drift to compute the surprise.

[3]We follow the convention of Hou, Xue, and Zhang (2020) and do not estimate a random walk model with drift to compute the surprise.

$$\text{REV}_t = \exp\left(\sum_{\tau=t-60}^{t-13} \log(1+r_\tau)\right) - 1$$

Profitability

CBOP - Cash-Based Operating Profitability

Following Ball et al. (2016), we measure cash-based operating profitability

$$\begin{aligned} CBOP_t =& (REVT_t - COGS_t - XSGA_t + XRD_t \\ & - \Delta RECT_t - \Delta INVT_t - \Delta XPP_t + \Delta DRC_t \\ & + \Delta DRLT_t + \Delta AP_t + \Delta XACC_t)/AT_t \end{aligned}$$

where REVT is total revenue, COGS are cost of goods sold, XSGA are selling, general, and administrative expenses, and XRD are R&D expenses. Moreover, Δ refers to one-year changes, which we measure for accounts receivable (RECT), inventory (INVT), prepaid expenses (XPP), deferred revenues (the sum of current, DRC, and long-term, DRLT), trade accounts payable (AP), and accrued expenses (XACC). We set the missing values of XRD and all missing changes to zero.

GPA - Gross Profits to Assets

Following Novy-Marx (2013), we measure gross profits to assets as total revenues (REVT) minus cost of goods sold (COGS) scaled by current total assets (AT), i.e.,

$$GPA_t = \frac{REVT_t - COGS_t}{AT_t}$$

O - Ohlson's O-score

Following Ohlson (1980), we measure O-score with the following linear relation

$$\begin{aligned} O_t =& -1.32 - 0.407 \cdot \log(AT_t) + 6.03 \cdot \frac{DLC_t + DLTT_t}{AT_t} \\ & - 1.43 \cdot \frac{ACT_t - LCT_t}{AT_t} + 0.076 \cdot \frac{LCT_t}{AT_t} \\ & - 1.72 \cdot \mathbf{1}_{LT_t > AT_t} - 2.37 \cdot \frac{NI_t}{AT_t} - 1.83 \cdot \frac{PI_t + DP_t}{LT_t} \\ & + 0.285 \cdot \mathbf{1}_{NI_t<0 \text{ and } NI_{t-12}<0} - 0.521 \cdot \frac{NI_t - NI_{t-12}}{|NI_t| + |NI_{t-12}|} \end{aligned}$$

where AT corresponds to total assets, DLC to short-term debt, DLTT to long-term debt, ACT to current assets, LCT to current liabilities, LT to total liabilities, PI to pretax income, DP to depreciation and amortization, and NI to net income. We winsorize all variables except for dummy variables at the first and 99th percentile of their respective distribution.

OPE - Operating Profits to Book Equity

Following Eugene F. Fama and French (2015), we measure operating profits to book equity as

$$OPE_t = \frac{REVT_t - COGS_t - XSGA_t - XINT_t}{BE_t}$$

where REVT corresponds to total revenues, COGS to cost of goods sold, XSGA to selling, general and administrative expenses, XINT to interest expenses, and BE to book equity. The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero. Moreover, missing values in COGS, XSGA, and XINT are set to zero.

ROA - Return on Assets

Following Balakrishnan, Bartov, and Faurel (2010), we measure return on assets as quarterly income before extraordinary items (IBQ) relative to lagged quarterly total assets (ATQ), i.e.,

$$ROA_q = \frac{IBQ_q}{ATQ_{q-1}}$$

Finally, the earnings announcement date has to be after the fiscal quarter end date.

ROE - Return on Equity

Following Hou, Xue, and Zhang (2014), we measure return on equity as quarterly income before extraordinary items (IBQ) relative to lagged book equity (BEQ)

$$ROE_q = \frac{IBQ_q}{BEQ_{q-1}}$$

Quarterly book equity (BEQ) is computed as the (a) book equity of shareholders plus (b) balance sheet deferred taxes and investment tax credit minus (c) the book value of preferred stock. Regarding (a), we measure book equity of shareholders by shareholders' equity (SEQQ), or the sum of common equity (CEQQ) and the par value of preferred stock (PSTKQ), or if all previous items are unavailable by total assets (ATQ) minus total liabilities (LTQ). Regarding (b), the balance sheet deferred taxes and investment tax credit is TXDITCQ, or TXDBQ, or zero if both are missing. Regarding (c), the book value of preferred stocks corresponds to PSTKRQ, or PSTKQ, or zero if both are unavailable. Finally, the earnings announcement date has to be after the fiscal quarter end date.

TBI - Taxable Income to Book Income

Following Lev and Nissim (2004), we measure taxable income to book income as pretax income (PI) scaled by net income (NI)

$$TBI_t = \frac{PI_t}{NI_t}$$

We require positive pretax and net income.

Size

ME - Size

Following Eugene F. Fama and French (1992), we measure size as the natural logarithm of the market equity. We obtain market equity data from CRSP by multiplying the shares outstanding (shrout * 1,000, as they are given in thousands) with the corresponding share price (the absolute value of altprc), i.e.,

$$ME_t = \log\left(shrout_t * 1000 * |altprc_t|\right)$$

Trading frictions

AMI - Amihud Illiquidity Measure

Following Amihud (2002), we measure the illiquidity from daily return-to-volume (RV), which is the absolute daily return (r) scaled by the daily dollar trading volume (product of vol and prc), i.e.,

$$RV_t = \frac{|r_d|}{prc_t * vol_t}$$

Then, the Amihud illiquidity measure corresponds to the average return to volume estimate over the last six months. We require at least 50 observations for this average and adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

BETA - Beta Relative to the Market

Following Eugene F. Fama and MacBeth (1973), we measure the market beta β_1 from monthly CRSP data. Specifically, we run the following time series regression over the previous five years

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + u_t$$

Moreover, we require at least 24 monthly observations for the regression above.

DTV - Dollar Trading Volume

Following Brennan, Chordia, and Subrahmanyam (1998), we measure dollar trading volume from daily CRSP data as the average dollar trading volume (product of vol and prc) from month $t-6$ to month $t-1$ (with at least 50 observations). Moreover, we adjust dollar trading volume from NASDAQ according to Gao and Ritter (2010).

ISKEW - Idiosyncratic Skewness Relative to the Fama and French (1993) three-factor model

Following Bali, Engle, and Murray (2016), we regress daily excess returns in each month on the three Eugene F. Fama and French (1993) factor model (with a minimum of 15 daily observations), i.e.,

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

Then, idiosyncratic skewness is measured as the skewness of residuals u_t.

IVOL - Idiosyncratic Volatility Relative to the Fama and French (1993) model

Following Ang et al. (2006), we compute idiosyncratic volatility relative to the Eugene F. Fama and French (1993) three-factor model as the volatility of residuals from the following regression (with a minimum of 15 daily observations)

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

MDR - Maximum Daily Return

Following Bali, Cakici, and Whitelaw (2011), we measure the maximum daily (excess) return in each month from daily CRSP data (with at least 15 return observations).

TUR - Share Turnover

Following Datar, Naik, and Radcliffe (1998), we measure the daily share turnover of each stock as the number of shares traded (vol) scaled by the number of shares outstanding (shrout), i.e.,

$$T_t = \frac{vol}{shrout}$$

Then, the share turnover for each firm in each month t is the average daily share turnover over the previous six months (with at least 50 observations). Moreover, we adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

Valuation

BM - Book Equity to Market Equity

Following Davis, Fama, and French (2000), we measure the book-to-market ratio as book equity (BE) from Compustat divided by market equity (ME) from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$BM_t = \frac{BE_t}{ME_t}$$

The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero.

CFM - Cash Flow to Market Equity

Following Lakonishok, Shleifer, and Vishny (1994), we measure the cash-flow-to-price ratio as income before extraordinary items (IB) plus depreciation (DP) scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$CFM_t = \frac{IB_t + DP_t}{ME_t}$$

We exclude all stocks with negative cash flows.

DM - Debt to Market Equity

Following Bhandari (1988), we measure the debt-to-market ratio as short-term debt (DLC) plus long-term debt (DLTT) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$DM_t = \frac{DLC_t + DLTT_t}{ME_t}$$

We exclude stocks with missing DLC and DLTT observations.

EBM - Enterprise Book Equity to Market Equity

Following Penman, Richardson, and Tuna (2007), we measure enterprise book equity scaled by market equity as net debt plus book equity scaled by net debt plus market equity

$$EBM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + BE_t}{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments, and ME to the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD). Lastly, in this case, book equity BE is computed as common equity (CEQ) plus TSTKP minus DVPA. Missing observations in DVPA and TSTKP are set to zero. Additionally, we require that the sum of net debt and book equity, as well as the sum of net debt plus market equity, are positive.

EM - Earnings to Market Equity

Following Basu (1983), we measure the earnings-to-price ratio as income before extraordinary items (IB) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$EM_t = \frac{IB_t}{ME_t}$$

We exclude firms with negative earnings.

NDM - Net Debt to Market Equity

Following Penman, Richardson, and Tuna (2007), we measure net debt to price as net debt from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$NDM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t}{ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments. Lastly, missing observations in DVPA and TSTKP are set to zero.

NPY - Net Payout Yield

Following Boudoukh et al. (2007), we measure the net payout yield as net payouts from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$\begin{aligned} NPY_t = &[(DVC_t + PRSTKC_t + \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV<0}) \\ &- (SSTK_t - \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV>0})]/ME_t \end{aligned}$$

where DVC are dividends from common stock, PRSTKC is the purchase of common and preferred stock, PSTKRV is the value of the net number of preferred stocks outstanding, and SSTK reflects the sale of common and preferred stocks. $1_{\Delta PSTKRV<0}$ is a dummy variable that has value one if the annual change in PSTKRV is negative and zero otherwise. Moreover, we exclude stocks with negative net payouts.

OCM - Operating Cash Flow to Market Equity

Following Desai, Rajgopal, and Venkatachalam (2004), we measure the ratio of operating cash flows to price as operating cash flows (OC) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$OCM_t = \frac{OC_t}{ME_t}$$

Before 1988, we measure OC as funds from operations (FOPT) minus the change in working capital (item WCAP). Thereafter, we measure OC as net cash flows from operating activities (OANCF). Moreover, we exclude firms with negative operating cash flows.

SM - Sales to Market Equity

Following Barbee Jr, Mukherji, and Raines (1996), we measure the sales-to-price ratio (SM) as sales (SALE) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$SM_t = \frac{SALE_t}{ME_t}$$